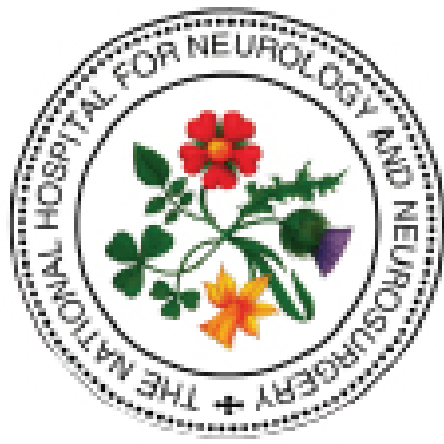




BLOOD GLUCOSE and INSULIN THERAPY in PATIENTS with ACUTE BRAIN PATHOLOGY

AIM to AVOID HYPO / HYPERGLYCAEMIA and LARGE VARIATIONS in GLUCOSE LEVELS

Hyperglycaemia is a common secondary insult in TBI, SAH and ACUTE STROKE and is associated with poor neurological outcome. Some patients will develop 'stress hyperglycaemia', others will have a prior history of diabetes or previously undiagnosed diabetes. Recommendations for glycaemia control in the acute phase of brain injury are the same for patients with or without prior history of diabetes.

Tight BG control only required if diagnosis involves acute brain pathology – for all other patients follow UCLH protocol (BG target 4-10mmol/L).

ACUTE PHASE OF BRAIN INJURY Blood Glucose Target 7-12mmol/L	
If BG >12mmol/L start IV insulin infusion 50u Human Actrapid in 50mL sodium chloride 0.9%	
Blood Glucose (BG) mmol/L	INITIAL Rate of Infusion units/hour
12.1 - 14.9	1
15 - 17.9	2
18 - 21	3
>21	4

Only use **SINGLE STRENGTH INSULIN**

This is **STARTING SCALE ONLY** – must be adjusted after BG check at 1 HOUR

People with **Type 1 Diabetes** need insulin at all times delivered SC or IV and must be referred to Diabetes CNS

BLOOD GLUCOSE MEASUREMENT

The sampling site may influence glucose measurements (*main sources of discrepancies are vasoconstriction, low blood flow rate, state of shock, ischemia, or oedema*)

Check BG readings on arterial blood and glucometer correlate – use arterial sample only when additionally checking blood gases, otherwise use glucometer

Refer known diabetic patients to Diabetes CNS

INSULIN

Insulin is the generally preferred form of treatment as dose can be titrated more rapidly than oral agents and does not have dose ceiling

Make small changes in IV insulin and check trend

When BG at lower target consider **reduced rate** rather than stopping in patients who have required insulin to avoid high / low swings

Maintain insulin infusion for 1 hour after starting SC insulin or oral anti-diabetic medicine

Stop insulin infusion when interrupting EN and/or TPN for **more than 2 hours**
if BG still high after that period, restart insulin infusion (possibly with glucose infusion background for diabetic patients)

If using glucose, rate should be at least 3g/h (i.e. 10% at 30mL/h or 5% at 60mL/h)

METFORMIN

For patients previously on metformin consider contraindications to continuing: *Renal impairment / Heart failure / Need for radiographic contrast studies*

Consider reintroducing pre-admission oral diabetic medication when BG stable – oral medication not normally withheld even if feed stopped for short period

DRUG RELATED FACTORS

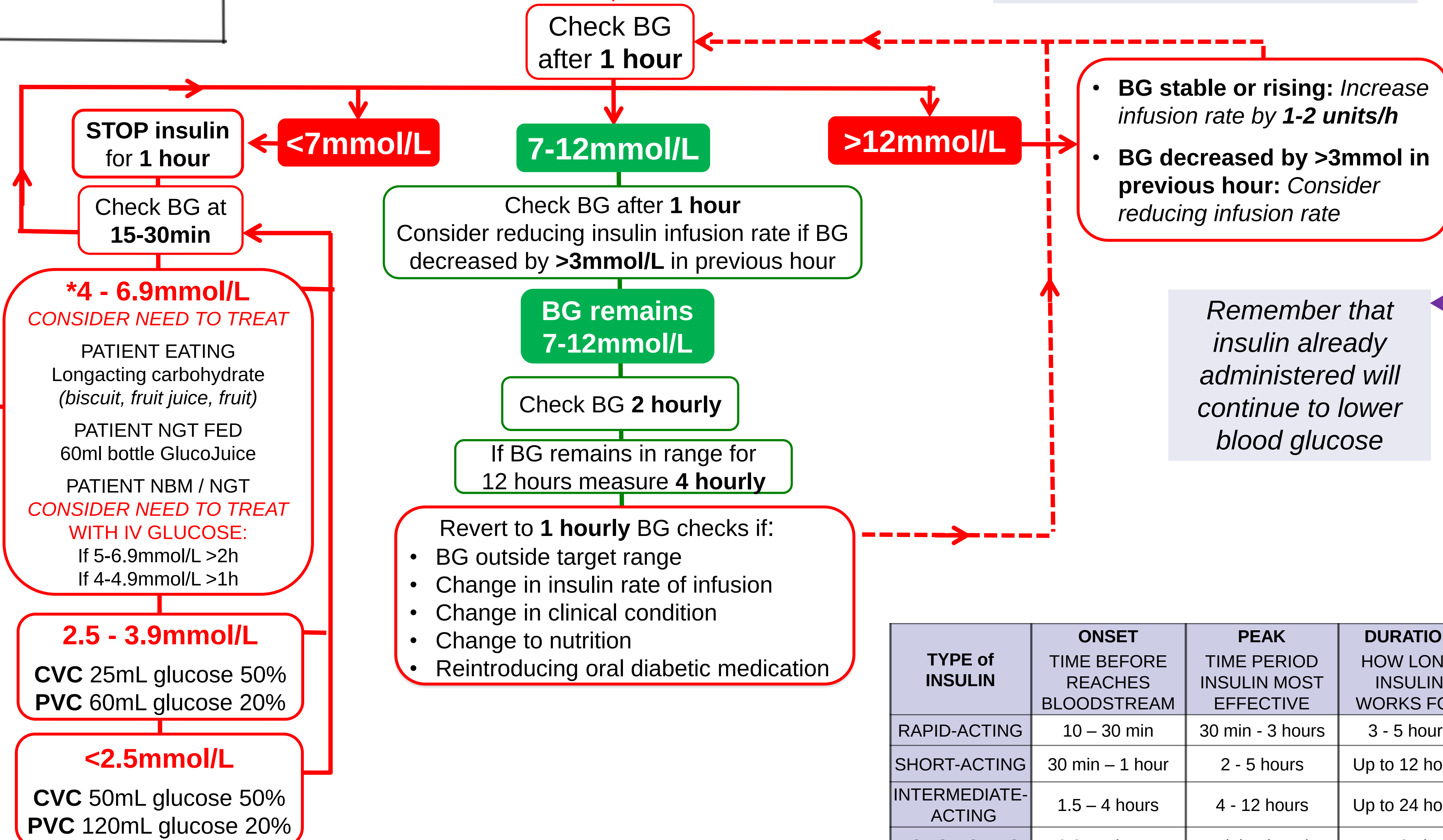
Insulin doses may need to be higher when certain drugs are co-administered (*steroids and vasopressors*)

DISCHARGE TO WARD

Blood Glucose Target 4–10mmol/L

Review drug chart

- Prescribe sliding scale for those patients who require continued insulin therapy
- Patients on insulin infusion will have IV glucose + potassium infusion prescribed to run concurrently
- Consider whether this should be started at least 1 day prior to discharge to ward
- Dietician to change feeding regime



TYPE of INSULIN	ONSET TIME BEFORE REACHES BLOODSTREAM	PEAK TIME PERIOD INSULIN MOST EFFECTIVE	DURATION HOW LONG INSULIN WORKS FOR
RAPID-ACTING	10 – 30 min	30 min - 3 hours	3 - 5 hours
SHORT-ACTING	30 min – 1 hour	2 - 5 hours	Up to 12 hours
INTERMEDIATE-ACTING	1.5 – 4 hours	4 - 12 hours	Up to 24 hours
LONG-ACTING	0.8 – 4 hours	Minimal peak	Up to 24 hours

